

Software Engineering Practice Midterm

1. The following code violates two major style conventions. What are they, and how would you change the code to fix them?

```
public class questionThree {  
  
    private int MyVariable = 0;  
  
    public int getVariable() {  
        return MyVariable;  
    }  
}
```

2. Consider the following unit test that wants to check the functionality of the Widget class. Does the unit test currently check that functionality, and if not, how would you change the test to fix that?

```
public static interface Database {  
    int doAThing();  
}  
  
public static class Widget {  
    private Database db;  
  
    public Widget(Database db) {  
        this.db = db;  
    }  
  
    public boolean makeWidget() {  
        int val = db.doAThing();  
        return val > 0;  
    }  
}  
  
@Test  
public void testWidget() {  
    Database db = Mockito.mock(Database.class);  
    Widget testWidget = Mockito.mock(Widget.class);  
    Assert.assertEquals(false, testWidget.makeWidget());  
}
```

3. Consider the following code:

```
public class UsbDriveReader {  
    private final UsbDrive drive;
```

```

    public UsbDriveReader(UsbDrive drive) {
        this.drive = drive;
    }
    public String readData() {
        String result = "";
        while (!drive.fullyRead()) {
            try {
                result += drive.readChar();
            } catch (IOException e) {
                // we've read as much as we can, break out of the
loop
                break;
            }
        }
        return result;
    }
}

public interface UsbDrive {
    boolean fullyRead();
    char readChar() throws IOException;
}

```

Suppose you wanted to test the error handling in the UsbDriveReader. Write a unit test that will accomplish that - you can use any libraries we've covered in class, and you do not need to write the import lines for the code (assume the IDE will handle those).

4. Consider the following code. If you needed to return an arbitrarily large number of strings, what would be a different method signature (return type + method name + parameters) that you could use instead of `List<String> loadStrings()`?

```
public static interface LoadManyStrings {  
  
    List<String> loadStrings();  
  
}
```

5. Suppose you are trying to implement the following code, and need to fix the TODO. What methods/parameters/return values/etc could you add to `CatFacts` to help solve the TODO? Note: it's ok if your solution would require code changes in other classes outside `CatFacts`, for this question you just need to write the code that you would add to the `CatFacts` class.

```
public static interface CatInfo {}  
  
public static class CatFacts {  
  
    public void newFact(CatInfo info) {  
        // TODO: share this cat fact with several publishers and many people  
    }  
}  
  
public static interface BookPublisher {  
    /** Collects cat facts into an anthology to publish */  
    public void addCatInfoToAnthology(CatInfo info);  
}  
  
public static interface Person {  
    /** Read the wikipedia article about the new cat fact*/  
    public void readWikipedia(CatInfo info);  
}
```

6. Suppose you want to build some software that allows one user to upload a large CSV file for processing and distribution. The workflow that you want to support is that first, a user uploads a file and a list of email addresses. Next, the server will let the user know if the file and email addresses are valid. If they are, then finally the server will then process the CSV file and email out the results to the provided addresses. Draw a System Design Diagram for this workflow.

7. Consider the following API. What methods/return values/parameters would you change (and to what) to improve the flexibility of the API for future changes?

```
public static interface LoginService {  
    public boolean login(LoginRequest request);  
  
    public boolean logout(LoginRequest request);  
}
```